



Transferring Appearance with Mask-Guided Attention*

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Abstract

Despite significant advancements in large-scale text-to-image generation and text-conditioned image editing, appearance transfer remains relatively unexplored. Transferring appearance aims to transfer an object's appearance in an appearance image to an object in the structure image so that background details are preserved and accurately reflect the transferred object's characteristics. Appearance transfer has practical applications in areas like virtual try-on and e-commerce product placement. Existing methods often require fine-tuning text-to-image diffusion models or are not applicable to virtual try-on and e-commerce scenarios, which is not ideal. In this paper, we introduce a Mask-Guided attention mechanism that replaces the existing self-attention in U-net architecture of Stable diffusion [29]. This approach can be easily integrated into the Masactrl [6] framework, enabling appearance transfer without model fine-tuning and suitable for a wide range of applications. Our method uses masks of objects in images to guide the appearance transfer process, with these masks obtained from the Segment Anything Model (SAM) [17]. This integration of SAM-generated masks allows for precise object localization and more accurate appearance transfer. We have conducted comprehensive experiments on transferring various clothing items (shirts, jeans, t-shirts) onto people, as well as transferring sofas into living spaces.

CCS Concepts

• Computing methodologies → Image processing; Computer vision.

Keywords

Appearance transfer, image editing, virtual try-on, sofa placement, masked attention, segmentation.

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1 Introduction

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The objective of appearance transfer is to transfer the appearance of an object from an appearance image to an object in a structure image by consistently preserving the details of the object in the appearance image and the background in the structure image. An illustrative example would be transferring the appearance of a pink doughnut to a grey one as depicted in Figure 1. The process involves two key steps to achieve this successfully: first, identifying and linking semantically equivalent areas between the two doughnuts and then faithfully transferring the pink doughnut's appearance through these connections without compromising the grey doughnut's fundamental structure. This technique has wide-ranging applications, including virtual try-on experiences and furniture placement visualization. Previous attempts in this field have typically operated under the assumption that appearance transfer occurs between objects with similar shapes [8, 21] or requires training [32] or are not applicable for the use-cases of virtual try-on and sofa placement applications [8, 10, 19, 21, 36].

Early GAN-based works focused primarily on virtual try-on of clothing on human models in limited contexts or controlled environment [3, 15, 18]. Other approaches relied on expensive 3D augmented reality (AR) or virtual reality (VR) technologies for items like furniture in rooms [2, 33], making them impractical for scaling to the billions of products that often lack 3D models.

Through the study of recent diffusion models, studies have found that the denoising network's self-attention and cross-attention components are able to grasp significant semantic details from the images they produce [6, 13, 37]. Building on the functions of the queries, keys, and values within these self-attention layers, our key insight is to employ the self-attention mechanism across different images, which we term Mask-Guided Attention.

In this paper, we introduce Mask-Guided Attention, which replaces the existing self-attention in the Diffusion U-net layers. It can be incorporated into Masactrl [6] to smoothly transfer the appearance. Specifically, we first invert a structure image using Null-Text Inversion [20], resulting in the diffusion process (DP1) for the structure image synthesis. In the next diffusion process (DP2), we invert an appearance image, resulting in the diffusion process (DP2) for the output image synthesis. In the final diffusion process (DP3) of editing the structure image, we can use the Query features in DP3 self-attention to query the corresponding Key and Value features in DP1 self-attention for the foreground and DP2 self-attention for the background. In other words, we transform the existing self-attention into 'Mask-Guided Attention', where the attention operation happens in the self-attention of three related diffusion processes. All of the existing works [1, 36] on appearance transfer either model to preserve the background in structure or appearance image but not both. But our model can be easily adapted



Figure 1: Results of applying mask-guided attention for the transfer of appearance from appearance image to structure image.

to preserve either of the backgrounds, which will be evident from our formulation.

2 Related Work

2.1 Appearance Transfer

The process of appearance transfer can be viewed as a specific type of image-to-image translation. In contrast to Neural Style Transfer [9], which focuses on transferring an overall artistic style between images, our focus lies in semantic style transfer. This entails transferring the appearance of semantically related regions between two images.

Earlier techniques have used generative approaches like Generative Adversarial Networks (GANs) [11], to train on large collections of paired or unpaired images. Swapping Autoencoders [25] encoded structure and appearance separately, but required a dedicated generator and large dataset for each target domain. To minimize supervision, some approaches utilized a single reference image [34]. Tumanyan et al. [36] trained a dedicated generator for each image pair, leveraging a pre-trained DINOViT [7] to extract and incorporate structural and stylistic information during the training process. However, this approach still necessitated training a generator for every image pair.

Recent studies have explored the utilization of large-scale diffusion models for transferring visual appearances without the need for additional inputs or model training [1, 8, 21]. These methods commonly use loss functions on the latent codes, guiding the denoising process to produce images that maintain a reference image’s structure while changing its visual features. Training-free approaches based on self-attention modification include Eye-for-an-eye [10], explicitly rearranges the appearance feature map to match the spatial arrangement of semantics in the structure image, and Ctrl-X [19], employs feature injection and spatially-aware normalization in the attention layers for appearance transfer.

In contrast to the aforementioned approaches, our method operates without any training or per-image optimization, respects the semantic correspondences between the two images during appearance transfer, and is applicable to a wide range of use-cases.

2.2 Image Editing with Diffusion Models

Expanding on recent progress in large-scale diffusion models [23, 26–28, 30, 31], various studies have explored innovative approaches to enhance control over the generative process. By leveraging these models, researchers have developed techniques for editing tasks [4, 5, 13, 16, 20, 22, 24, 37, 39–41] that allow users to better manipulate the editing process. Certain studies have indicated that adjusting the denoising network’s internal representations, specifically its attention layers, can be effective for image editing [1, 6, 13, 37].

A recent project, MasaCtrl [6], revealed that by freezing the keys and values of the self-attention layers during non-rigid image edits, the original appearance of the image is better preserved. Our approach involves injecting keys and values into the self-attention layers, but injection occurs between different images rather than between an image and its output edit.

3 Preliminaries

3.1 Latent Diffusion Model

Diffusion models [14, 35] are generative models that can generate desired data samples from Gaussian noise by gradually removing the noise. A diffusion model defines a fixed forward process and a learnable reverse process. The forward process adds the noise to the data sample $\mathbf{x}_0$ to generate the noisy sample $\mathbf{x}_t$ with a predefined noise adding schedule α_t at time-step t :

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}(1 - \bar{\alpha}_t)\mathbf{I}) \quad (1)$$

where $\bar{\alpha} = \prod_{i=1}^t \alpha_i$. After T steps, the data sample $\mathbf{x}_0$ is transformed into Gaussian noise $\mathbf{x}_T \in \mathcal{N}(0, \mathbf{I})$. The reverse process tries to remove the noise and generate a clean sample $\mathbf{x}_{t-1}$ from the previous sample $\mathbf{x}_t$:

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \sigma_t) \quad (2)$$

where μ_{θ} and σ_t are the corresponding mean and variance, respectively. The variance is a time-dependent constant, and the mean $\mu_{\theta}(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \epsilon \frac{1 - \alpha_t}{1 - \bar{\alpha}_t})$ can be solved by using a neural network $\epsilon_{\theta}(\mathbf{x}_t, t)$ to predict the noise ϵ . To train such a noise estimation

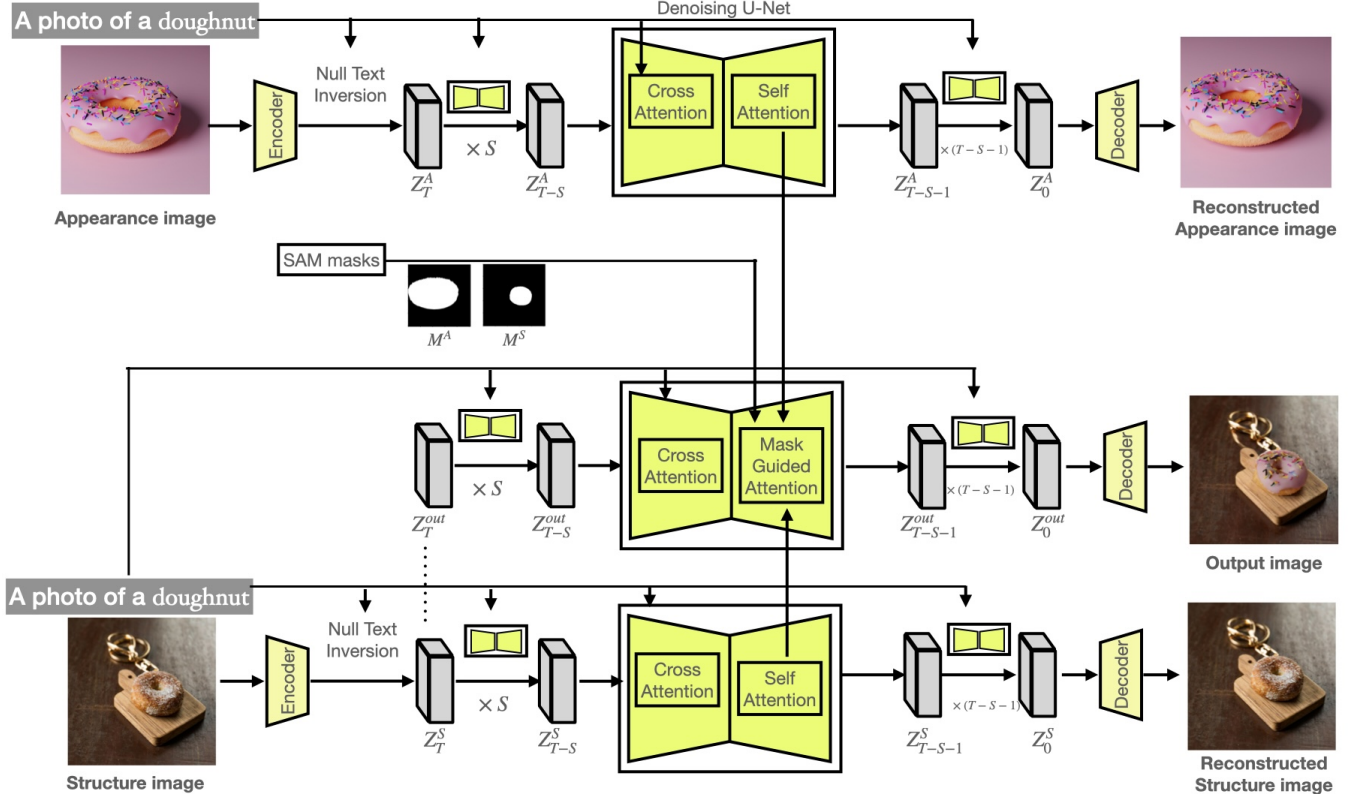


Figure 2: Pipeline of the three diffusion processes (DPs). The top DP reconstructs the I^A . The bottom DP reconstructs the I^S . The middle one is the DP that incorporates Mask-Guided Attention to generate I^{out} .

network ϵ_θ , the objective is the simplified mean-squared error:

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0, \epsilon, t} \left(\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2 \right). \quad (3)$$

By sampling $\mathbf{x}_{t-1}$ iteratively, a data sample $\mathbf{x}_0$ can be generated from random Gaussian noise $\mathbf{x}_T$. In addition, text prompts P can be conditioned into the predicted noise $\epsilon_\theta(\mathbf{x}_t, t, P)$ so that the diffusion models can generate text-aligned images.

3.2 Attention Mechanism in Stable Diffusion

The denoising U-Net ϵ_θ in the Stable Diffusion model is an encoder-decoder architecture consisting of a series of basic blocks, and each basic block contains a residual block [12], a self-attention module, and a cross-attention [38] module. At each timestep t of the denoising process, the noised latent code z_t is given as input to the denoising U-Net network. The intermediate features of the network at layer ℓ , denoted $\phi_\ell(z_t)$, are first projected into queries $Q = f_Q(\phi_\ell(z_t))$, keys $K = f_K(\phi_\ell(z_t))$, and values $V = f_V(\phi_\ell(z_t))$ through learned linear projections f_Q, f_K, f_V .

For every query vector $q_{i,j}$ positioned at spatial location (i, j) of Q , we calculate an affinity score, or attention score, with respect to all keys in K , reflecting how relevant each key is to the corresponding query. These attention scores are then normalized using a softmax operation, defining the weight each value will have when

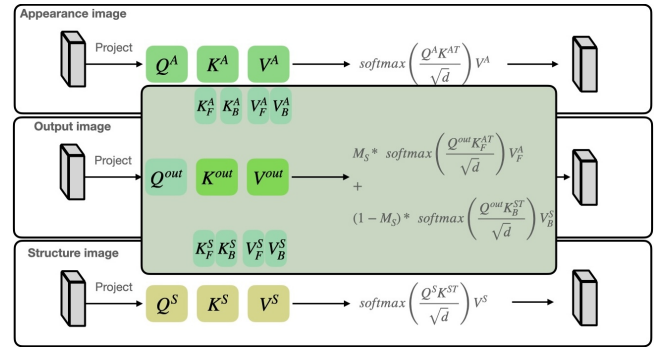


Figure 3: Mask-Guided Attention.

updating the features at position (i, j) . Finally, the weighted values are aggregated to produce the output for each query position. Formally,

$$\Delta\phi(i, j) = \text{softmax} \left(\frac{q(i, j) \cdot K^T}{\sqrt{d}} \right) \cdot V \quad (4)$$

where $\Delta\phi(i, j)$ denotes the aggregated output feature at (i, j) used to update the spatial features $\phi(z_t)$.

4 Method

Given a pair of images (I^S, I^A), we wish to generate an output image I^{out} depicting the structure of the subject present in I^S with the appearance of the subject in I^A by preserving either the background of I^S or I^A . In a recent work, [6] explored the self-attention layers within the denoising network of a text-to-image diffusion model in the perspective of performing the non-rigid edit on the object present in the image. They show that keeping the keys and values of these self-attention layers fixed aids in preserving the visual characteristics of objects when applying non-rigid manipulations over a given image.

Masactrl [6] contains two diffusion sampling processes, DP1 and DP2. DP1 starts with latent noise z_T obtained from inversion and corresponds to the reconstruction of the original image. DP2 is the sampling process of the edited image that corresponds to the new prompt. By starting DP2 with the z_T of DP1 and the edited prompt, the resulting image captures the structure mentioned in the edited prompt (for e.g., running dog, sitting dog, etc.), but we cannot expect the object in the result to follow the semantics of object in the original image. Since DP1 captures the characteristics of the object in the original image, they incorporate information from DP1 into the DP2 process by replacing self-attention in DP2 with mutual self-attention.

Building on this insight, we observe that one can utilize the queries, keys, and values from these self-attention layers to transfer semantic information between different images. The queries identify the semantic significance of each spatial location. The keys then provide context for each query, enabling the model to gauge the significance of various image parts for a specific query position. Finally, the values represent the content we aim to generate and define the information that will be used for determining the features of each query position.

Mutual self-attention, defined in [6], allows each query to attend to all the keys in the original image sampling process. This could confuse the edited image sampling process for the object region as it also attends to the background of the original image. Moreover, the object doesn't need to attend to the background. So, we have ensured that the object region only attends to an object by considering the object masks. Object masks obtained from cross-attention are inaccurate, accounting for the poor quality of the output image. We cannot also obtain the mask of a region at any granularity required for us from cross-attention (for ex., the mask of the shirt in the image of a person). To circumvent this problem of finding the mask of the appropriate region, we substituted this process by using the Segment Anything Model [17], which can accurately find the masks at any granularity.

4.1 Mask Guided Attention

Given the images I^S and I^A and their prompts P^S and P^A , we begin by inverting them using null-text inversion introduced in [20], which produces the latents z^S and z^T . We obtain the foreground object region masks M^S and M^A corresponding to I^S and I^A using the Segment Anything Model [17]. After obtaining the inverted latents, denoted by z^S and z^A , we perform three parallel diffusion processes (DP). DP1 starts with latent z^A and prompt P^A , resulting in the reconstruction of I^A . DP2 starts with latent z^S and prompt

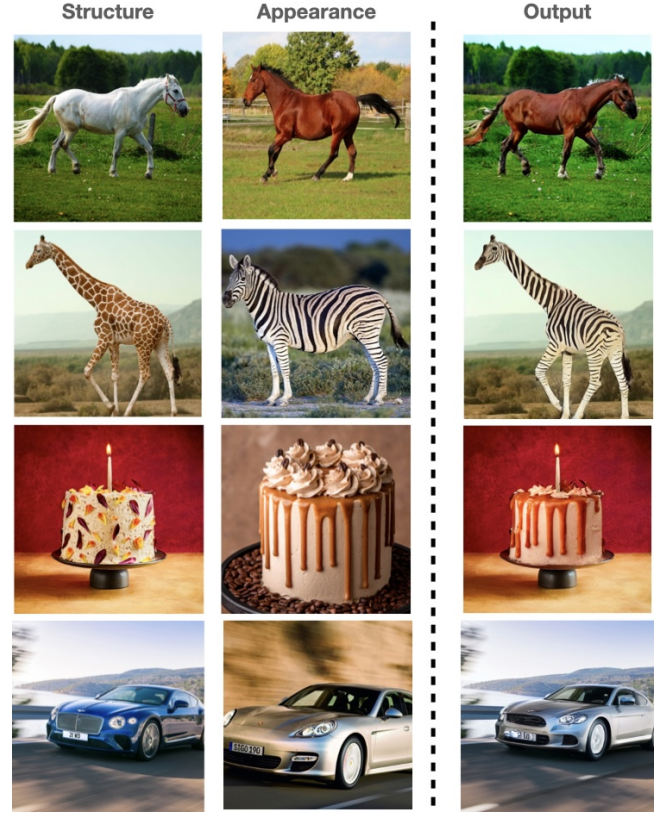


Figure 4: Results of applying Mask-Guided Attention on generic dataset.

P^S , resulting in the reconstruction of I^S . DP3 starts with latent z^S and prompt P^S with Mask-Guided Attention replacing the self-attention in Diffusion U-Net layers resulting in the output image I^{out} . The overview of all three processes is depicted in Figure 2.

At each timestep t and layer l , Q^S, K^S and V^S represent query, keys and values after projection in DP2. K_F^S and K_B^S represent the keys corresponding to the object region and background region of I^S . V_F^S and V_B^S represent the values corresponding to the object region and background region of I^S . The same representation goes for the queries, keys and values of I^A . Q^{out}, K^{out} and V^{out} represent query, keys and values after projection in DP3.

$$\begin{aligned}
 \Delta\phi^S &= \text{softmax}\left(\frac{Q^S \cdot K^{ST}}{\sqrt{d}}\right) V^S \\
 \Delta\phi^A &= \text{softmax}\left(\frac{Q^A \cdot K^{AT}}{\sqrt{d}}\right) V^A \\
 \Delta\phi_o^{out} &= \text{softmax}\left(\frac{Q^{out} \cdot K_F^{ST}}{\sqrt{d}}\right) V_F^A \\
 \Delta\phi_b^{out} &= \text{softmax}\left(\frac{Q^{out} \cdot K_B^{ST}}{\sqrt{d}}\right) V_B^S \\
 \Delta\phi^{out} &= M^S * \Delta\phi_o^{out} + (1 - M^S) \Delta\phi_b^{out}
 \end{aligned} \tag{5}$$



Figure 5: Results of applying Mask-Guided Attention on T-shirts of humans.



Figure 6: Results of applying Mask-Guided Attention on pants of humans.

Specifically, mask-guided attention in DP3 as described in Eq:5, queries corresponding to object region in Q^{out} attends on keys and values corresponding to the object region of I^A , i.e., K_F^A and V_F^A . Queries corresponding to the background region in Q^{out} attend on keys and values corresponding to the background region of I^S , i.e. K_B^S and V_B^S . Mask-guided attention is only applied within the decoder layers of the U-Net. $\Delta\phi^S$, $\Delta\phi^A$ and $\Delta\phi^{out}$ corresponds to intermediate features of the diffusion processes DP1, DP2 and DP3 respectively. Considering object masks, we restricted the object region of I^{out} to attend only the object region of I^A and rest only on the background region of I^S . Mask-Guided Attention is depicted in Figure 3.

5 Experiments

We apply the proposed method to the state-of-the-art text-to-image Stable Diffusion [29] model v1.5 with publicly available checkpoints. We obtain object masks from the publicly available SAM model. During sampling, we perform DDIM sampling [35] with 50 denoising steps, and the classifier-free guidance is set to 7.5. Extensive experiments were performed on the Animal Faces HQ (AFHQ) dataset and flowers dataset. We created our own dataset, namely, *Wild-Pairs*, which included 20 images from different domains taken from the internet. As an application for the use-case of sofa placement, we collected several two-seater and three-seater sofa images from Amazon website. Several images of shirts, jeans, and tee shirts were also collected from Myntra to showcase the generalizability of the model.



Figure 7: Results of applying Mask-Guided Attention on two-seater sofa.

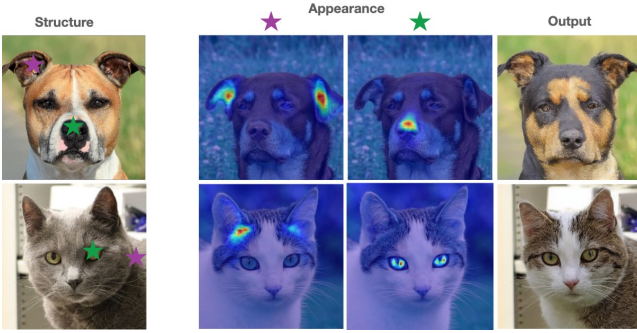


Figure 8: Aggregated self-attention of specific regions in the output diffusion process.

All the images are of resolution 512×512 . In Figure 4, we present the results on generic images. Results on the use cases of virtual try-on and sofa placement are depicted in Figures 5, 6 and 7. The same idea of mask-guided attention can also be tweaked to attend to appropriate regions to change the background, as shown in Figure 10. Our model gives better results because each pixel in the object region of the output diffusion process attends to relevant regions of the object in the appearance image, as depicted in Figure 8. Source code is accessible at <https://github.com/mani-312/Mask-Guided-Attention>.

5.1 Evaluation metrics and Comparison With Previous Work

For evaluating structure preservation and appearance alignment quantitatively, We report two metrics: DINO Self-sim measures self-similarity distance [7] in the DINO-ViT [36] feature space to assess structure preservation; DINO CLS evaluates the loss between DINO-ViT global [CLS] tokens to measure appearance alignment. For evaluating background preservation we reported LPIPS [42] between background regions of structure and output images assesses perceptual image similarity using a VGG network. We collected a dataset comprising images of dogs, cats, horses, and flowers, with 20 pairs of structure and appearance images in each category. Table 1 and Figure 12 compare Mask-Guided-Attention to the baseline. Our model yields superior results compared to existing approaches for background preservation. Our results are comparable to training-based methods like SpliceViT [1] and consistently outperform training-free methods [1, 10, 19]. The use-case of virtual try-on and sofa placement demands the preservation of the background of the structure, which the existing works fail to do as depicted in Figure 9.

5.2 Ablation Study

For better preservation of the background of the structure image, we tried different attention approaches, such as replacing K_B^S and V_B^S with K_B^{out} and V_B^{out} in mask-guided attention. The other way we

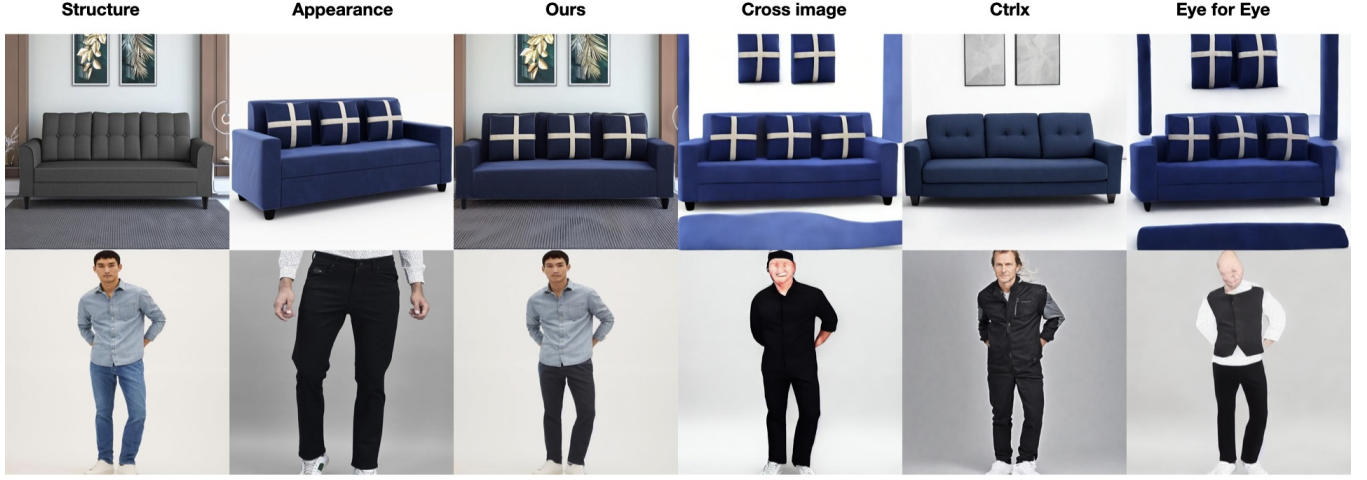


Figure 9: Existing methods cannot apply to virtual-tryon and sofa placement use cases.

Table 1: Comparison of different models on various metrics.

Metric \ Model	Ours	Splice[36]	Cross Image[1]	Ctrl-X[19]	Eye for Eye[10]
LPIPS	0.086	0.155	0.283	0.323	0.295
Self-sim	0.042	0.043	0.065	0.060	0.066
DINO CLS	0.075	0.040	0.046	0.110	0.035



Figure 10: Results of applying Mask-Guided Attention for changing the background of structure image with background image.

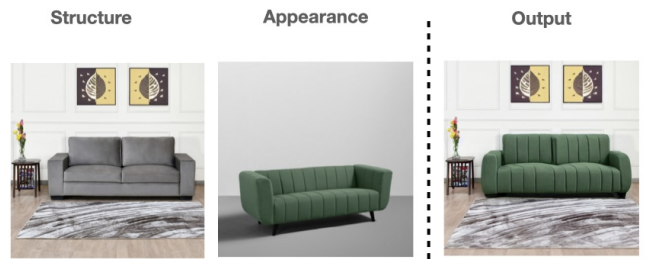


Figure 11: Our model fail to correlate output structure with appearance image.

tried was to concatenate both of them as $\{K_B^{out}, K_B^S\}$ and $\{V_B^{out}, V_B^S\}$. Our results depicted in Figure 13 conclude that these ablations could not preserve the background as much as the proposed mask-guided attention does.

5.3 Limitations

Our problem statement clearly states that the output image preserves the structure of the object in the structure image. However, when it comes to the use case of sofa placement, sometimes the structure of the sofa in the structure and appearance images may not be aligned. In such cases, our model struggles to get the structure of an object in the appearance image. One of those failure cases is depicted in Figure 11. The same goes for the virtual try-on. Our model struggles to change the jeans pants to cargo pants. This shortcoming may be circumvented by focusing on the aspect of

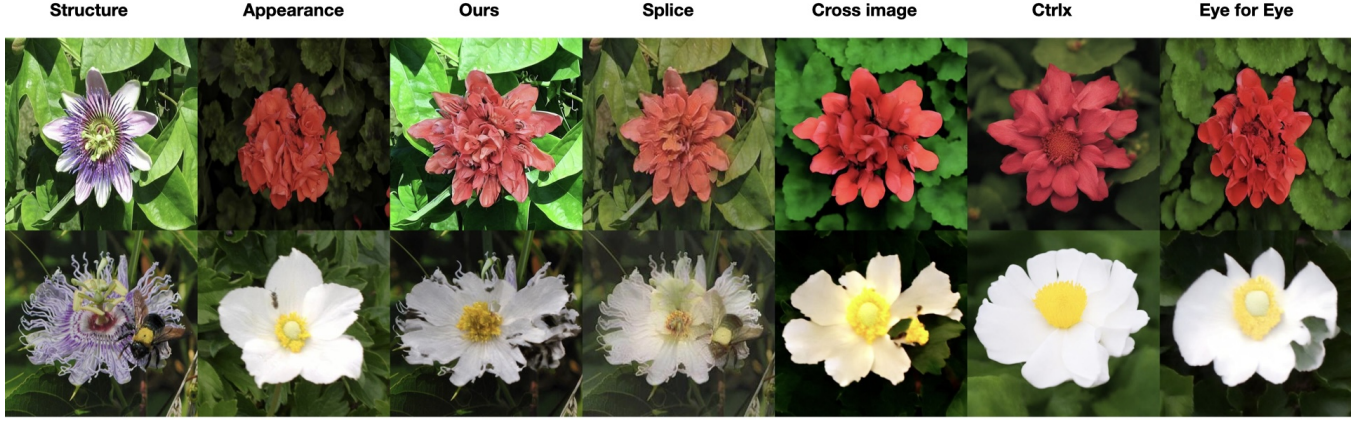


Figure 12: Our model outperforms others in structure preservation, appearance alignment and background preservation.

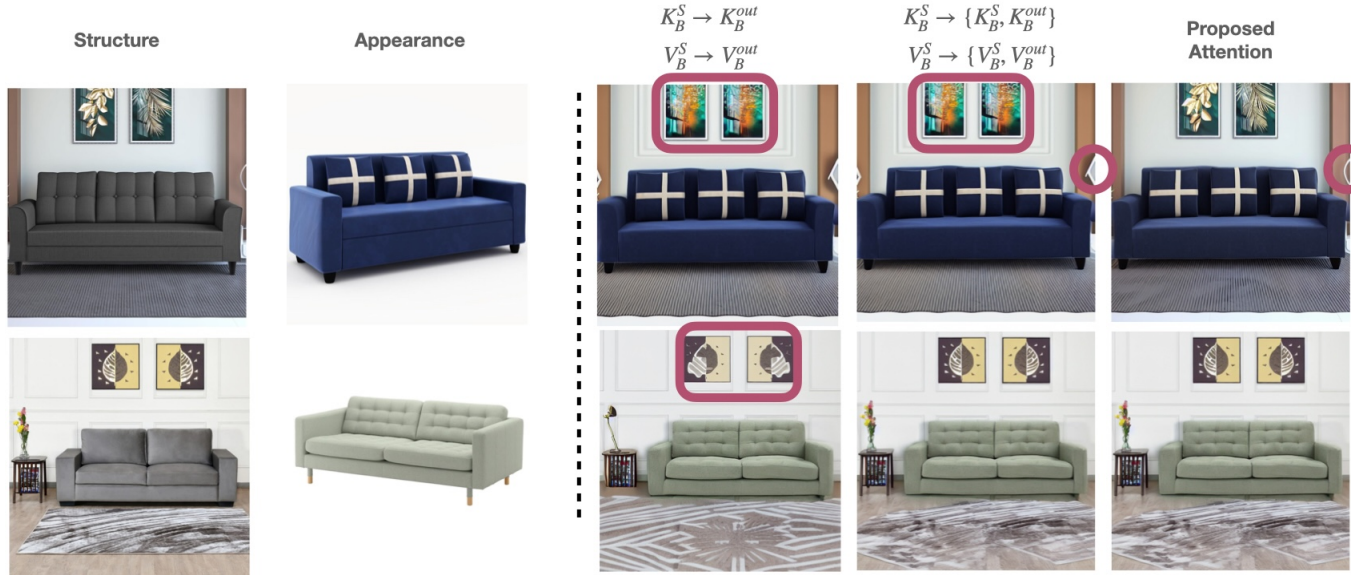


Figure 13: Ablation study on variations of Mask-Guided Attention.

each pixel attending to all the pixels, which could sometimes miss out on the important region. The other way to ensure the structure of the appearance object is being preserved is to take some weighted combination of z_t^A and z_t^{out} in a meticulous manner at each timestep. This could be the direction of future work to pursue better structure preservation.

6 Conclusion

We introduced a novel technique called mask-guided attention, which enables the seamless transfer of semantic attributes between objects in natural images. This is achieved by guiding the attention-based masks generated by the Semantic Affinity Model (SAM). Notably, our approach accomplishes this transfer without requiring any model training or user-provided conditioning. Moreover, this transfer capability can be applied to diverse applications, such as

sofa placement and virtual try-on scenarios. By examining the components of the self-attention layers, including the queries, keys, and values, we discovered that they inherently establish semantic correspondences between objects by blending these elements across different images. Leveraging this mask-guided attention mechanism, our method can produce gradual and realistic appearance transfers, resulting in high-quality, visually compelling images.

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